# Molecular Shapes (VSEPR)

Electron domains, shapes, bond angles and polarity

Name _____ Class _____ Date _____

**ELECTRON DOMAINS**

2 linear 180° · 3 trigonal planar 120° · 4 tetrahedral 109.5°

5 trigonal bipyramidal 90° and 120° · 6 octahedral 90°

A double or triple bond counts as one domain

Draw the Lewis structure first, then count domains on the central atom.

## A. Understanding

**1.** State the main idea of VSEPR theory.

**2.** Explain the difference between the electron-domain geometry and the molecular shape, using water as an example.

## B. Predicting shapes

**3.** For each molecule, give the number of electron domains and lone pairs on the central atom, the molecular shape and the approximate bond angle: (a) $BCl_3$ (b) $H_2S$ (c) $PCl_3$ (d) $CS_2$ (e) $SiH_4$

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**4.** Predict the shape around the carbon atom in formaldehyde, $CH_2O$, and the approximate H–C–H bond angle.

**5.** Predict the shape of the ammonium ion, $NH_4^+$.

## C. Bond angles

**6.** The bond angles in $CH_4$, $NH_3$ and $H_2O$ are 109.5°, about 107° and about 104.5°. Explain this trend.

## D. Polarity

**7.** State whether each molecule is polar or nonpolar, with a reason: (a) $CO_2$ (b) $H_2O$ (c) $CCl_4$ (d) $NH_3$

## E. Challenge

**8.** $XeF_4$ has six electron domains on xenon, two of which are lone pairs. Predict its molecular shape and explain where the lone pairs go.

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**9.** $SF_4$ has five electron domains on sulfur, one of which is a lone pair. Name its shape.

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